# Bohemiancoin

Today's financial systems — and unfortunately many cryptocurrencies as well — often become opaque, bloat to enormous proportions, and favour those with the greatest capital. Bohemiancoin v2 was born out of the need to change this direction. It returns control to the ordinary user, but with a clear message: real freedom requires firm order and personal responsibility.

The system does not promise effortless miracles, but mathematical fairness. Instead of endlessly growing data, Bohemiancoin is anchored in a fixed, 512 MB Codex. You do not need giant data centres to participate in the network. Your digital property does not reside on foreign servers, but exclusively in your pocket in the form of CBOR receipts. This allows you to securely store your assets even in a completely offline environment and cryptographically prove their existence and origin to anyone. With this absolute control, however, comes responsibility for your own data — the network's history lives in users' pockets, not in the cloud.

The competitive struggle of Hunters (miners) is transformed by The Cubic Seal algorithm from mindless electricity burning into elegant geometric precision. And thanks to the 15:10:5 transaction rule, the network ignores no one. Those who are in a hurry can pay more, but the system always mandatorily processes transactions from those who wait patiently in the queue.

Bohemiancoin is a tool for those who want to have their finances one hundred percent under control, respect immutable rules, and are willing to accept the responsibility that true independence brings.

# Codex

The Codex does not store transaction history (who sent what to whom a year ago). Only the final result is stored. The Codex (512 MB) functions as a State Database. The Block itself does not represent a new link in a chain, but a so-called Codex Delta — a set of instructions that change specific values in this database. If a User wants to send an Actum, they must present proof that their request matches what is recorded in the current Codex. After successful verification in the Virbitirium, an atomic operation is performed in the Codex: the amount is deducted from the User and credited to the recipient. If either of these steps fails, neither is performed.

Because every node (computer in the network) has an identical copy of these 512 MB at any given moment, consensus in the network is very easily verifiable. Anyone can verify in a fraction of a second whether their version of the Codex is authentic, or whether someone is trying to deceive them. The Codex will have a clean 512 MB exclusively for user data. The Header is a separate part of the Codex but is still part of it. The Header is 64 MB. If we need to add new features in the future (such as voting on network development), there will always be room in the 64 MB for this.

### Header Composition

**Section A:** Current Epoch (approx. 16 MB): Every block is recorded here minute by minute. The hashes of the last 2.1 million blocks are here. This is the 'hot zone'.

**Section B:** Master Hash Registry (approx. 32 MB): Only the final fingerprints (Master Hashes) of all past epochs are stored here. One Master Hash (32 bytes) represents four full years of history.

**Section C**: System Data (remainder): Network rules, difficulty setting R, etc.

| Component | Size | Function | Backup |
|---|---|---|---|
| Header | 64 MB (Fixed) | Backbone, Master Hashes, Verification. | Backed up by every user. |
| Codex | 512 MB (Fixed) | Live network state, balances, Labyrinth. | After 4 years it is 'frozen' as an Archive. |
| CBOR | A few KB | Your private receipt, your proof. | Stored on your mobile/PC. |

# Transaction Flow

A User wants to send 10 GRS. The user's wallet performs a 'small time test' (a computational task) tied to the received Chronos Ticket. At the moment a node sends the User data about their slot (balance, Nonce), it attaches a Chronos Ticket — the current network time and the hash of the last block. This test takes 3–5 seconds, which naturally slows down mass spamming of the queue of the '10 oldest'.

Their wallet looks at the current Codex (which the User holds locally, or to which they are connected). Using Hash Mapping, it immediately finds their slot at a specific byte within those 512 MB. To allow a node to immediately find a user's slot in the 512 MB file without searching, we use Direct Map Hash. See the technical sheet for details. It verifies that there are sufficient funds and the correct Nonce (to ensure the transaction is unique). The User creates an Actum (a signed command). At that moment, in their local copy of the Codex (and subsequently across the entire network after broadcasting), their 10 GRS moves into the Grey Zone. Technically: the status bit in their 64-byte slot is changed to Pendens. These funds are 'locked'. The User cannot send them again, but the recipient does not yet have them.

# Virbitirium (Mempool)

The Actum travels to the Virbitirium. Here it waits for validation. From this point, no one can tamper with it, as it is cryptographically signed by the User.

### Master Hash Epoch (Closing)

Once the network's internal time elapses, the process begins:

Validation: All Acta in the Virbitirium that passed the check are definitively recorded. Final Write: In the Codex, 10 GRS is deleted from the User and 10 GRS is added to the recipient, and a New Codex is created: The old Codex is 'stamped' into a Master Hash. This Master Hash is written into the Header of the new Codex. The Header retains the history of the 4

most recent old Codexi for continuity and security. New coins are released (a reward for the system/validators).

### Batching & Geometric Continuity (Efficiency)

Instead of chaos and wasted energy, Bohemiancoin uses geometric order. Fixed Manifest: Each block contains exactly 30 transactions. This creates a predictable network throughput. Train Scheduling: If the Virbitirium (waiting room) is under pressure (e.g. 120 transactions), Hunters divide them into 'blocks'. Alex seals block 101, John immediately follows with block 102. No Wasted Work: Thanks to the Cubic Seal and the fixed Codex, John's node can instantly recalculate its attack vector toward target T, even if Alex has just changed the state of the Codex. If their transactions do not overlap, John's block is accepted a second after Alex's. This Manifest is essentially an instruction package (Delta). The network does not transmit the entire Codex, only these differential changes.

## Chronos (Time in Bohemiancoin)

Chronos (internal time) is a combination of three elements that fit together like gears in an astronomical clock.

### Chronos: Composition of Internal Time

So that User A in New York and User B in Tokyo see the same thing, Chronos in every Header will look like this: Master Hash Epoch: A large time segment (e.g. 1024 Codexi). After an Aevum ends, new coins are released and a major archiving takes place.

Codex Sequence (Page Number): The ordinal number of the current 512 MB block within the given Epoch (e.g. Codex #42 in Epoch #5). Actum Index (Network Heartbeat): The ordinal number of the last recorded Actum in the current Codex. Example Chronos notation: 5 : 42 : 10502 (Epoch 5, Codex 42, transaction number 10,502). This code is a unique timestamp that cannot repeat.

### Chronos Ticket

Chronos is not merely a passive timer, but an active tool for asset protection. When a user creates a transaction, they receive a so-called Chronos Ticket from the network. This ticket contains the exact current block number and defines the Logical Expiry (e.g. valid for 12 blocks). The ticket is firmly bound to the transaction (Actum) and serves as a mathematical guarantee: if the transaction is not recorded in the network in time, its validity automatically expires.

### Codex vs. Master Hash Epoch

The Codex is closed frequently (e.g. every minute or after reaching a certain transaction capacity). Process: The Virbitirium is 'poured' into the 512 MB data section. Hash: A hash of only this data is created and written into the Header. Continuity: The new Codex references the hash of the immediately preceding one. State: Both Bob and Alex have identical 512 MB.

## Master Hash Epoch (Grand Finale)

This process occurs after a certain number of Codexi (e.g. once a day or after 1,000 Codexi). Seal: The hash of the last Codex is taken and subjected to 'Master' validation. Emission: Only at this moment does Chronos 'advance' to the new Epoch number and the network generates new Bohemiancoins. Archiving: The old Codexi from this Epoch can be 'stored' and users take their history with them for future audits.

## Fixed Header & Archiving (Immortality)

This is the greatest safeguard. The system does not bloat, but 'crystallises'. Constant 64 MB Header: This is the backbone of the network. It contains the IDs of all blocks and Master Hashes of all Epochs. Its size is always 64 MB — never more, never less. It is a lightweight, instantly distributable 'anchor' that everyone has on their mobile.

Header Backup: The Header is also backed up together with the Codex. Each user therefore keeps 'the list of all laws' (Header) and 'the state of all accounts' (Codex). Master Hash Epoch (4 years): At the end of each epoch, the 512 MB Codex is 'frozen' and confirmed by the Master Hash in the Header.

## Why is this good for synchronisation between User A and User B?

With the Actum Index (transaction sequence number), there can be no dispute about what came first. If User A sends a transaction with index 10502 and Alex sends another at the same moment, the validators in the Virbitirium assign them a fixed order. Once this order is written into the Codex, it becomes part of the Chronos. User A and User B then simply compare their Headers: 'Do you have Chronos 5:42:10502? Yes? Then we have exactly the same zeros and ones in our 512 MB.' When User A creates an Actum, their funds enter the grey zone within the current Codex. But what if the Codex closes before their transaction is validated? The grey zone is carried over to the next Codex (the funds remain blocked).

## Persistent Grey Zone Mechanism (Persistent Pendens)

For the grey zone to survive the closing of one Codex and continue into the next, we use the Status Bit in the user's 64-byte slot. Slot Write: When User A sends 10 GRS, the flag for 10 GRS in their slot in the current Codex is switched to Status: Pendens. State Transfer: When the Codex is closed and a new one is generated (transition from Codex #42 to #43), this Pendens state is copied. These 10 GRS remain unavailable to User A in the new Codex. Funds in the Grey Zone (Pendens) are protected by the Logical Expiry of Chronos. If the Actum is not written into the Manifest within a fixed window (e.g. 12 blocks), the lock is synchronously released across the entire Codex. This ensures that the user's assets never get stuck in a 'digital vacuum' and the network remains in perfect consensus without any central intervention.

Release/Write: The grey zone disappears only when: Validation OK: The amount is definitively deducted from Bob and credited to the recipient (precisely at the moment of the Master Hash Epoch or earlier according to the rules). Validation FAIL: If the Actum proves invalid, the Pendens flag is deleted in the next Codex and Bob's funds turn 'green' again (active). The Virbitirium is not merely a data repository. It is a decentralised queue with the following properties:

Immutability: Once an Actum enters the Virbitirium, no one (not even a miner or User A) can change it. If a single bit were changed, the digital signature (Seal) would no longer match the content and the network would immediately eject it. Transparency: Every network node can see what is waiting in the Virbitirium, ensuring that no one can jump the queue 'under the

table'. Independence from the Codex: The Virbitirium lives in its own memory layer so as not to slow down the lightning-fast reads from the 512 MB Codex.

## Entry Process (Gateway to the Virbitirium)

Before an Actum becomes part of the Virbitirium, it must pass through 'customs'. This customs check uses data from the Codex that User A has brought:

'Self-Audit' Validation: The node checks whether the user's Actum contains the correct reference to the current Chronos and Codex. Grey Zone Check: The node looks into the 512 MB Codex at the index of Bob's address. If Pendens status (grey zone) from another transaction is already showing there, this new Actum is rejected as an attempt at Double Spend. Cryptographic Test: It is verified that the signature matches the sender's public key.

When a node receives the user's first Actum, it looks at their 64-byte slot in the Codex. Available Balance: Total number of coins minus what is already in the Grey Zone from previous (not yet confirmed) transactions.

**Example: User A has 100 GRS.**
Actum 1 (20 GRS): The node sees 100 available. 20 GRS is immediately marked as Pendens (Grey Zone) in the Codex. 80 remain available.
Actum 2 (50 GRS): The node sees that 20 is 'in grey', so 80 are available. 50 GRS is added to the Grey Zone. 30 remain available.
Actum 3 (40 GRS): The node sees that 70 is in grey (20+50). Only 30 are available. Since it wants to send 40 GRS, this Actum is immediately rejected at the entrance to the Virbitirium.

## Nonce (Sequence Number)

So that User A cannot confuse the network by sending three identical transactions (e.g. 3x 20 GRS) and claiming it is just one, we use a Nonce in each slot. Each Actum must have a Nonce one higher than the previous one — for example, Actum 1 has Nonce 5 and Actum 2 must have Nonce 6. If User A were to send Actum 2 with the same Nonce as Actum 1, the Virbitirium would discard it as a duplicate.

### Multiple Transactions in the Grey Zone

All three Acta (if User A has sufficient balance) enter the Virbitirium. In the 512 MB Codex, the sum of all three shows under User A in the Grey Zone (Pendens state). User A cannot further manipulate them. Definitive Settlement (Block Seal): Once these Acta are included by a Hunter in the Manifest and the block is successfully sealed (The Cubic Seal), immediate settlement occurs. The Grey Zone for User A disappears, the funds are definitively deducted from them and immediately credited to the recipients. Carry-over to the Next Page (Safety Net): Only in the event that the Actum does not fit into the current block is its 'Pendens' state carried forward and it waits in the Virbitirium for the next Hunter.

### What if one of the transactions fails?

Deletion from the Virbitirium: The node detects an error (e.g. invalid signature or insufficient funds after previous transactions) and immediately deletes Actum 2 from the queue (Virbitirium). Unblocking in the Codex (Grey Zone Clearance): When creating the next Codex, the Pendens (Grey Zone) state for User A is automatically reduced by the amount (e.g. 50 BHC) that belonged to this invalid Actum. Return to User: These blocked funds are simply 'returned' to User A. Technically, this means that in their 64-byte slot they are no longer marked as 'grey' and are again 'green' (fully active and ready for the next use).

### Selection of Acta

Selection: The validator selects 30 transactions from the Virbitirium according to the 15:10:5 recipe.
15 with a fee: Sorts transactions by fee (GRS/byte) and takes the top 15.Wealthy users have priority (15 slots) if they are in a hurry. Patient users are guaranteed (10 slots) that they will always be served.The network itself has reserved space (5 slots) to live and evolve (Smart-Acta) — see the discussion at the end of the whitepaper.

### Announcement (Submitting the Ticket)

The validator creates a Manifest — essentially just a list of 30 IDs (hashes) of those transactions. The validator 'knocks' on the Codex module with their Manifest. Indexing: The Codex immediately (using fast jumps in 512 MB memory) verifies all 30 senders. Check: 'Does User A have sufficient GRS in slot #5000? Yes. Is sender B genuine? Yes.' Result: If even one transaction is erroneous, the Codex rejects the request. The validator must correct the Manifest.

# The Cubic Seal

### (Validation Mechanism)

Point T must be completely random, unpredictable, yet fair — it must emerge from a 'cocktail' of data that is mixed only a second before the start.

### Composition of Point T:

Hash of the Previous Block: This is the anchor. No one knows how the previous block will turn out until it is recorded. This gives the network continuity. Chronos (Network Time): The precise second in Bohemiancoin. Public Address of the Validator: This ensures that every validator in the world has their own unique Point T. Even if a neighbour could see your block, their Point T is in a completely different location in the cube. Manifest (Those 30 Transactions): This is the safeguard. If you wanted to change a single coin in the package, your Point T in the cube would immediately 'fly off' to different coordinates and your work would be worthless.

```
T = HashLastBlock + Chronos + Address + Manifest
```

Geometrisation: The resulting 256-bit hash is divided into three parts, which determine the coordinates xt, yt, zt in an imaginary cube. State: Point T is born. It is fixed, unpredictable, and belongs only to you. nstead of simply searching for random numbers, the Codex defines a digital universe in which the node must physically 'aim' at a target.

Geometric Law of the Codex: The Codex is not merely a passive database (512 MB), but an active creator of space. It defines an inconceivable range of coordinates 2^256 x 2^256 x 2^256, within which all 'Hunting' takes place.Coordinate System: Each data packet (Manifest, Address, Nonce) is interpreted by the Codex as a specific point x, y, z. The calculation is not random, but firmly defined by the Codex algorithm.

Virtual Reality of Data: The playing field does not physically occupy additional RAM, but is 'rendered' over the existing Codex data. To calculate the coordinates, the node must use data directly from the Codex slots (e.g. balances), which makes simulation impossible without holding the entire 512 MB block.

Decentralised Consensus: Because every node in the world runs on an identical Codex, all see the same universe boundaries and the same rules of movement.

### The 'Padding' Principle

If there are only 5 transactions in the network, the Hunter cannot simply 'lie'. To generate Point T, the packet must have a precise structure. The Hunter adds so-called Null-Transactions to the Manifest. These empty records are not random — they are system signatures of the network confirming that at that given second (Chronos) there truly was no more work. Result: Point T is born from 5 real transactions and 25 'system stamps'.

### Dynamic 'Curvature' of Space (Codex Protection)

This mechanism ensures that the digital universe is not static, but constantly changing, making it impossible to predict results.

Network Pulse and the Key to Curvature: With each new block, the old coordinates become invalid. The hash of the last block serves as the DNA for the current round. Higher-Level Anti-Farm: Codex Salt per-block: Hardware Salt binds the Codex to a specific machine, but an attacker could physically replicate the Codex onto multiple disks. Therefore, with each recorded block, an XOR operation is performed on the entire 512 MB Codex with a small pseudorandom masking vector derived from that block's hash. The result is that a copy of the Codex made 5 blocks ago no longer yields correct addresses for the Labyrinth in this round.
Unique Perspective: The validator's Public Address is mixed into the curvature formula. Each node sees the Target (Point T) at a completely different location in the universe.

### Process: The Cubic Seal

The entire process takes place in a digital space of dimensions 2^256 x 2^256 x 2^256.

### Step 1: Birth of the Origin (CPU)

Input: Hash of the last block + Hunter's Address + Manifest (30 transactions) + Hardware ID.
Action: Calculation of 100,000 chained hashes. Result: Starting coordinates *P0 x, y, z.*
Significance: A unique start for each Hunter. Hardware ID ensures that the computation cannot be transferred to another machine. Mathematical notation: Start_XYZ = Hash(PrevBlock + HunterID + HardwareID)

### Step 2: Vector Labyrinth (Disk + Bitrate)

To eliminate corporate farms with extremely fast RAM disks, the system uses a 1:1 binding to physical disk. Sequential Dependency Chain: Instead of simply measuring time, each jump in the Labyrinth (10,000 jumps of 4 KB each) becomes the input for the calculation of the next jump. Data from the disk is hashed with the obtained value and the result determines the address of the next jump:

```
addr_0 = BLAKE3(P_0 || HardwareID)
data_i = Codex.read(addr_i, 4096 B)
addr_{i+1} = BLAKE3(data_i || state_i) mod (512 MB / 4096)
```

This ensures that the resulting Point T cryptographically depends on actual data read from the physical disk layer by layer. Anyone who holds the data in RAM and does not honestly perform these cryptographic operations on a valid salted Codex will get a different Point T and therefore an invalid block.

### Step 3: The Seal Shot (GPU)

Hardware: Graphics card. Action: Generating random Nonces. Each Nonce creates a Point P in the cube. Mathematics: The winner is whoever finds a Point P whose Euclidean distance d to the target (Point T) is less than the allowed radius R:

*Victory: d = sqrt((xp - xt)^2 + (yp - yt)^2 + (zp - zt)^2)*

If d < R (current difficulty radius), the node has Sealed the block.

To maintain a stable pace (Target: 1 block / 60 seconds, i.e. 1,440 blocks/day), the radius R is dynamically adjusted based on a calibration window of 1,024 blocks (small Epoch).

```
R_new = R_old * ((1024 * 60) / SUM(times_of_1024_blocks_in_s))
```

This calculation is completely deterministic. The value R must not fall below R_min nor exceed R_max. All nodes arrive at the same value without voting. Times are recorded in Section A of the Header and publicly verifiable.

### Step 4: Validation and Recording (Network)

The winner broadcasts to the network their certificate (The Block), which contains among other things a Proof-of-Physical-Read (PoPR) — a set of 10 randomly selected verification points (Breadcrumbs) from their Labyrinth path.

Efficiency: Each Breadcrumb is only 32 B (hash) + 4 B (address). In total this adds a negligible 360 B per block. Verification Speed: Other nodes reproduce these points simply by reading data from their Codex (reading only 10 addresses x 4 KB = 40 KB of data). Verification takes under 10 ms. Fairness: Breadcrumbs are selected deterministically from the winning Nonce, so the attacker cannot choose in advance which positions they will need to prove to the network.If the shot fits (e.g. your d = 48 beats the competing d = 50) and the PoPR is valid, the 30 transactions are 'fired' into all Codexi and a new round begins.

## Anti-Farm Safeguards (Summary)

| Safeguard | Function | Target |
|---|---|---|
| Hardware Salt | The Codex is uniquely encrypted with the disk/CPU ID. | Prevents copying of data between machines in a farm. |
| Codex Salt per-block | The mask of the 512 MB Codex changes with each block. | Devalues static Codex backups across farms. |
| Sequential Dependency | The jump address depends on the hash of the previous read. | Eliminates software emulation of RAM latency (spoofing). |
| Proof-of-Physical-Read | Fast deterministic check of 10 jumps (Breadcrumbs). | Proves to the network that data passed through physical disk reading. |

## Hardware Salt v2: TPM Attestation (Optional Trust Tier)

The current Hardware Salt is based on a self-reported disk/CPU ID — the node tells the network what hardware it uses, and the network trusts it. This is functional, but a weak point. TPM Attestation addresses this weakness without raising the barrier to entry for ordinary users.

**What is TPM?**

Trusted Platform Module (TPM) is a cryptographic chip built as standard into modern PCs and laptops (mandatory for Windows 11). It contains a private key that physically cannot be extracted or copied to another machine. TPM can sign arbitrary data in a way that cryptographically proves the signature was created on this specific physical hardware.

**Optional Tier — Principle:**

**Tier 1 (Basic):** Nodes without TPM work normally with self-reported Hardware ID. No change, no entry barrier.

**Tier 2 (TPM):** When registering, the node has its TPM chip sign its public address. The resulting certificate is written once into the 64 MB Header. From that moment, the network knows that this Hunter runs on verified physical hardware.

**Tier 2 Advantage:** In the event of a collision between two blocks with the same distance d, the node with a TPM certificate has priority. A TPM node thus achieves statistically higher success without needing more powerful hardware.

**Farm Protection:**

An attacker may have a thousand machines, but the TPM key cannot be copied by software. Each node in the farm must undergo a separate physical registration. A Tier 2 farm of nodes is administratively costly — which is precisely the effect that Hardware Salt promises, but now mathematically guarantees.

**Consistency with Bohemiancoin's Philosophy:**

The system does not force anyone to use expensive equipment. TPM is today a standard component of an ordinary laptop — not a premium feature. At the same time, it naturally rewards those who invest in greater network trustworthiness. This is mathematical fairness in practice: the same rules for everyone, but more trust for those who prove more.

## Assembling the Seal Certificate (The Block)

Your node immediately creates a data package that serves as irrefutable proof. It contains: Hunter ID & HW Fingerprint: So the network knows who to send the reward to. Winning Nonce: This is the 'key' which, when inserted into your algorithm, yields the coordinates of Point P. 10 Breadcrumbs: Control coordinates from your Labyrinth path (e.g. position after every 1,000th jump). Manifest: Those 30 transactions you just 'locked'.

**Precision Rule: If your shot had distance d = 50 and User B had d = 48, their block won.**

### Broadcasting to the Network (Gossip Protocol)

The node sends this certificate to its nearest neighbours (Peers). They perform a lightning check: They take your HW ID and run your 10 Breadcrumbs through their Codex. They measure the distance d between your Point P (from the Nonce) and Point T (from the Breadcrumbs). If it matches, they pass the package on. Within approximately 200–500 ms, the entire planet knows of your victory.

### Rewriting Reality (The Codex Patch)

This is the moment when a 'virtual puzzle' becomes an 'accounting ledger'. Every node in the network (not just yours) must do the following: Open the 512 MB file: The program looks at the hash of your winning block. It tells it: 'Overwrite data at these 30 addresses in the Codex.' Write Transactions: The old data in the Codex (which formed the walls of the labyrinth) is overwritten with data from the new transactions (from User A to User B). New Version: Chronos on every node advances the version counter by +1. The old labyrinth ceases to exist at that second.

## Life Cycle of an Actum

### Birth (Initiation and Signing)

User Alex wants to send 100 Grošes (GRS) to Beatrice. Codex Query: Alex's wallet connects to a node. The node immediately finds Alex's slot using Direct Map Hash. Chronos Ticket: The node sends Alex the current data (balance, Nonce) and a Chronos Ticket (network time + hash of the last block). Anti-Spam Test: Alex's mobile/PC must perform a 3–5 second computation tied to this ticket. (This is the new protection.) Signing: Alex signs the package. The Actum now contains: Sender, Recipient, Amount, Timestamp from the Ticket, Fee, and Nonce.

### Entry into the Virbitirium and Grey Zone

Validation: The node checks the signature and especially whether the time in the Actum matches the issued Chronos Ticket (preventing time fraud). Pendens Activation: Once the node accepts the Actum, Pendens state is immediately activated in the 512 MB Codex for Alex. Technical detail: The status bit in the 64-byte slot is changed and 100 GRS is virtually moved to the 'blocked' field. At the same time, a block number when this block expires (calculated from Alex's Chronos Ticket) is written into the reserved space in the slot (Pendens_Expiry). Alex cannot spend the funds twice, but has the mathematical assurance that if the transaction does not go through, the network will automatically unblock the funds for them upon reaching the expiry block.

### Selection into the Manifest (15:10:5 Rule)

The Hunter assembles their Manifest (30 transactions):
15 slots: Highest fees (Market Priority).
10 slots: Longest waiting according to the time from the Chronos Ticket (Social Peace).
5 slots: System Smart-Acta / Notaries (Infrastructure). If empty, filled with additional 'waiting' transactions.

### Geometric Sealing (The Cubic Seal)

Labyrinth: The Hunter performs 10,000 jumps in the Codex (at speeds of 15–250 MB/s). Shot: The GPU searches for Point P near target T. Breadcrumbs: The Hunter includes 10 control points from their labyrinth path in the block for instant verification by others.

### Recording into the Codex (Crystallisation)

Once the block is accepted, an atomic write occurs. Offset Jump: Nodes jump directly to the bytes of Alex and Beatrice. Settlement: 100 TOL is definitively deducted from Alex; Beatrice is credited. Release: The Pendens state for Alex disappears. Drip: The Hunter is credited 6.6 GRS. The entire write takes milliseconds thanks to the fixed file structure. Master Hash Verification: Immediately after these changes, the node recalculates the hash of the entire 512 MB file and compares it with the value in the block header. If it matches, reality is 'crystallised'.

### Archiving and CBOR (Personal Proof)

Receipt: Alex saves a CBOR file with a Merkle proof. Immortality: After the end of the Epoch (4 years), everything is 'locked' into the 64 MB Header. Alex has the proof in their pocket, the network remains lean (still 512 MB).

**Summary: The Actum begins as a signature, waits as a frozen state (Pendens), battles in the geometric labyrinth, and ends as an atomic write in the Codex and a CBOR receipt in the user's pocket.**

### Restart and Clearing the 'Grey Zone'

Automatic Release (Time-out): If the Actum does not fit into any of the subsequent blocks (e.g. due to a low fee) and the validity of its Chronos Ticket expires (e.g. 12 blocks have passed since its creation), a so-called Logical Dissolution occurs. All network nodes look at the current block number in the Header, compare it with the transaction's expiry, and synchronously cancel the Pendens flag in the 512 MB Codex. Bob's funds are immediately 'green' (active) again without any central command for deletion.

## Master Hash Epoch: The Grand Settlement

The Master Hash Epoch is not merely a recording of transactions, it is a cryptographic stamp on an entire era of reality. It takes place on two levels: Meso (medium) for network cleanup and Macro (large) for the economy.

### Network Timekeeping (Chronos)

No Clocks: The Epoch is not governed by real time (New York vs Prague), but by the number of writes (Block Height).
Small Epoch (Sync): Every 1,024 blocks (approx. 17 hours).
Large Epoch (Harvest): Every 2,100,000 blocks (approx. 4 years).

### Closing Process: 'The Grand Strike'

Once the counter reaches the limit of 2.1 million blocks, Grand Hunt mode is triggered:

Cubic Seal for Master Hash: Hunters stop searching for targets for 30 transactions and focus on a single fixed Master Point T, which is the result of hashing all 2.1 million previous blocks.

Extreme Difficulty: The strike radius R is drastically reduced. It is a global race for the 'Key to the Era'. Dynamic Target (Safety Net): If no one hits within the first 1,440 blocks (approx. 24h), the radius R begins to grow by 1% every hour until someone succeeds. This guarantees the network will not halt. Precision Rule: If two Hunters hit simultaneously, the winner with the reward is determined by the smaller distance d.

### User Receipt (CBOR) — Audit in Your Own Pocket

Once the block is sealed (your Step 4), the transactions are fired into the 512 MB Codex. At this moment, however, the network must not hold history forever. CBOR Generation: Bob's and Alex's wallets automatically download a miniature file in .cbor format (Concise Binary Object Representation) from the network. This is a super-fast and extremely data-efficient binary format. What's inside: This CBOR does not contain the entire network history. It contains only:

The hash of their specific transaction (Actum).
Merkle Proof (Cryptographic proof that this transaction is genuinely recorded in the current Codex).
Timestamp and signatures.

Result (Off-chain Proof): The network can comfortably 'forget' their transaction and overwrite it in the next round. But if in the future a tax authority or auditor wants proof of a payment, Alex simply shows them their .cbor file. Chronos will instantly verify it against the current Master Hash and confirm: 'Yes, this is a valid imprint of reality.'

## Life Cycle of a CBOR (Creation and Distribution)

Creation (At the Hunter's): A second after the Hunter solves the Cubic Seal, they generate a human-readable CBOR package from the transaction data. Sealing: A fingerprint (Merkle Root) is calculated from this CBOR, which is written into the 64 MB Header. This anchors the receipt to the block. Distribution (Attachment): The Hunter sends to the network a package: [Header] + [State Delta] + [CBOR Attachment]. Enforcement: Other nodes accept the block only if the CBOR exactly matches the changes in the Codex.

## CBOR Roles

| Role | Action | Storage |
|---|---|---|
| Hunter | Verify and delete. Checks consistency and discards the CBOR. | Fixed 576 MB (512+64). |
| Archiver | Store forever. Builds a historical library of all CBORs. | Unlimited (grows over time). |
| Wallet | Back up. Downloads their piece and sends it to email/cloud. | Minimal. |

# Currency System

The currency is built on a solid foundation of whole numbers to prevent errors and inflation.

Main unit: 1 Groš (GRS)
Sub-unit: 1 Tolar (TOL)
Conversion ratio: 1 GRS = 1,000,000 TOL
Total supply: 42,000,000 GRS (42 trillion TOL). This amount is final and mathematically immutable.

### Currency Release

Coin emission does not occur continuously as with Bitcoin, but within three Large Epochs. Each Epoch lasts exactly 2,100,000 blocks (approx. 4 years).

### Emission Distribution (14,000,000 GRS per Epoch):

Hunter Reward (1%): 140,000 GRS is immediately obtained by the validator who first hits the 'Key to the Era' (Master Hash) using the Cubic Seal algorithm. Public Circulation (99%): 13,860,000 GRS moves into the network's reserve fund. Drip Circulation (The Drip): These 99% are released linearly with each subsequent block over the following 4 years.

Calculation: Each 1 block releases exactly 6.6 GRS (6,600,000 TOL) into circulation.

### Network Security: The Cubic Seal & Chronos

The network uses a unique 3D geometric Proof-of-Work that requires cooperation between CPU, GPU, and Disk.

Codex: Each node holds a 512 MB file (Labyrinth).
Bitrate Corridor: Read speed from disk must be between 250–3,000 MB/s. This makes mining on RAM disks or giant server farms impossible.

Precision Rule: In the event of a collision (two blocks simultaneously), the winner is the one who hit Point T with greater geometric precision (smaller distance d).
Atomic Write: Each block (30 transactions) is written 'all or nothing'. A power outage never damages the integrity of the Codex.

# Master Hash Epoch (Grand Finale)

Once the block counter reaches a multiple of 2,100,000, the network switches to The Grand Strike mode:

Internal Time: The network ignores earthly clocks. The Epoch is governed only by the number of confirmed blocks.
Expanding Target: If the 'Key to the Era' is not found within 1,440 blocks (approx. 24h), the target radius R begins to grow by 1% every hour until someone succeeds.

Normal Operation:
Hunters collect transaction fees (Fees) from users in each block (30 transactions).
Lifetime Shot: Once every 4 years they compete for 1% of new emission (140,000 GRS) when closing a Large Epoch.This system divides the network into Muscles (performance), Memory (archive), and Proofs (user). Everything is held together by the CBOR format.

## The 64 MB Header and Master Hash Principle (Auditability)

Here is the solution against bloating after 100 years:

Epoch (4 years): Every 2.1 million blocks, the history is 'smelted' into a single Master Hash (32 bytes).
Codex Reset: The 512 MB file is cleared, with only current balances remaining.
Fixed Header (64 MB): Old hashes of individual blocks are deleted. The Epoch's Master Hash is written into the Master Registry (a fixed field in the Header). Result: The Header is always 64 MB. Even after 100 years, it will contain only 25 rows with Master Hashes, taking up practically nothing.

### Audit

Even in 2126 you can prove a payment from 2026 as follows:
You retrieve your cbor file (from email/cloud). It contains a 'mathematical bridge' (Merkle Proof).
The current Hunter software looks in its 64 MB Header at the Master Hash Epoch 1 slot.
It compares your CBOR with this Master Hash.
Match = 100% proof of authenticity.

**Bohemiancoin does not need history in the network, because history is in the users' pockets and the mathematical 'anchor' (Master Hash) is in every node's 64 MB Header.**

CBOR in your pocket: Every user holds their digital receipts (CBOR) locally. Even offline you have physical proof of your assets, which you can show to anyone (Bluetooth/QR).

Lightning Return (Sync): If you return online after a week, you do not download the entire history. You download only the Delta (list of changes) and your node performs an Atomic Write

to specific addresses in the 512 MB Codex. You are synchronised within a second — see the technical sheet.

Mesh-Net Ready: Thanks to the small size of the Codex (512 MB), people can exchange the 'world map' locally even during a global network outage.

## Archiving: Digital Fossils (History)

After each Epoch (4 years), the system 'cleanses' itself to remain fast, but history remains public and immutable.

Master Hash: At the end of the Epoch, a single final fingerprint of the entire 512 MB Codex is generated. It is written into the indestructible 64 MB Header. Freezing: The old Codex is saved as a static file (e.g. Epoch_2026.bin). Nothing will ever be overwritten in it again. Democratic Backup: Anyone can download all past Epochs (e.g. 10 Epochs = 5 GB, fitting on an inexpensive flash drive). Verification: Anyone can verify the authenticity of an old backup simply by comparing its hash with the Master Hash in their 'live' application.

### Fixed Header & Archiving (Immortality)

This is the greatest safeguard. The system does not bloat, but 'crystallises'.

Constant 64 MB Header: This is the backbone of the network. It contains the IDs of all blocks and Master Hashes of all Epochs. Its size is always 64 MB — never more, never less. It is a lightweight, instantly distributable 'anchor' that everyone has on their mobile.

Header Backup: The Header is also backed up together with the Codex. Each user therefore keeps 'the list of all laws' (Header) and 'the state of all accounts' (Codex).Master Hash Epoch (4 years): At the end of each epoch, the 512 MB Codex is 'frozen' and confirmed by the Master Hash in the Header.

## Sub-section: 'Notarial Consensus' Module (Smart-Acta) — Idea. Not included in the prototype.

This supplement defines a mechanism by which verified Hunters in the Bohemiancoin network can close automated contracts (analogous to Smart Contracts), without disrupting the fixed data structure of the Codex.

### Qualification and Privilege (Merit-Based)

*The right to write a Smart-Actum into the system is not automatic. It is a privilege earned through demonstrable work for the network.*

*Entry Condition: The Hunter must have recorded at least 10 successfully sealed blocks in the current Epoch (verifiable in the 64 MB Header).*
*Maintaining Status: If the Hunter does not drop below the threshold of 10 blocks in the last window (e.g. 100,000 blocks), they maintain the status of 'Notary'.*
*Loss of Privilege: Upon attempting to validate a logically erroneous contract or upon a drop in activity, the node automatically loses the right to use the 5 system slots for Smart-Acta.*

### Execution Mechanism (Notarial Model)

Unlike Ethereum, the code does not run on all nodes but is delegated to Hunter-Notaries.

Initiation: The user approaches a Hunter-Notary with a contract request.
Recording: The Hunter uses one of the 5 system slots in the Manifest. They insert the hash of the conditions and an instruction for Persistent Pendens into the slot.
Collateral (Escrow): Coins designated for the contract are immediately switched to Pendens (Grey Zone) state in the Codex. They are frozen and await a mathematical trigger.
Automatic Payment: Once the parameters are met (e.g. Chronos time elapsed or the other party's signature), the funds in the nearest block are automatically released to the target address in the Codex.

### Security and Integrity

Logical Validation: Other Hunters, upon receiving a block, do not check the entire contract code, but only the correctness of the result (State Transition). If the result does not match the recorded rules, the block is rejected.
Fixed Impact: A Smart-Actum in the Codex occupies only the space of a standard transaction. All complexity lies in the Header and archives, thereby preserving the 512 MB size.

### Glossary

Actum: Transaction. A record that User A is sending a certain amount to User B.

Virbitirium: Mempool (waiting room). The place where all sent Acta gather and await processing. Codex: Distributed ledger (Ledger / Blockchain). The final, indisputable, and immutable state of all balances shared by the entire network.

# Technical Specification of Data Structures

### User Slot in the Codex (64 Bytes)

Each user is assigned a fixed slot of 64 bytes in the 512 MB Codex. The slot address is deterministically derived from the public key using the capacity of one Codex, which is fixed (8,388,608 slots).

Direct Map Hash: offset = SHA3-256(pubkey) mod (512 MB / 64 B).

Slot distribution is uniform due to the cryptographic property of SHA3.

**Structure of the 64-byte slot:**

```
[ 0.. 7]   Balance_GRS       uint64   (8 B)  — main balance in GRS
[ 8..15]   Balance_TOL       uint64   (8 B)  — sub-balance in TOL
[16..23]   Pendens_Amount    uint64   (8 B)  — amount in grey zone
```

```
[24..27]   Pendens_Expiry    uint32   (4 B)  — block number when Pendens
expires
[28..35]   Nonce             uint64   (8 B)  — transaction counter
[36..37]   Status_Flags      uint16   (2 B)  — bits:
           Active|Pendens|Frozen|TPM
[38..63]   Reserved          bytes    (26 B) — reserved for future
extensions
```

Codex Capacity: 512 MB / 64 B = 8,388,608 slots (maximum number of accounts in the network). This capacity is fixed and immutable. The scaling mechanism beyond this limit is described in the document Expansion of Bohemiancoin.

## Cryptographic Primitives

The following table defines the cryptographic algorithms used in Bohemiancoin:

Signatures (Seal): Ed25519 — 32-byte public key, 64-byte signature. Chosen for verification speed (batch verify ~50,000 signatures/s on a standard CPU), resistance to side-channel attacks, and compact key size.

Hash (all internal purposes): BLAKE3 — faster than SHA-256 even on CPUs without hardware acceleration, natively parallel, 256-bit output. Used for all internal hashes: slot offset, Merkle proof, Master Hash, Point T.

Merkle Tree (CBOR proof): Binary Merkle tree over 30 Manifest transactions. Tree depth is fixed (log2(30) ≈ 5 levels). The CBOR proof therefore always contains exactly 5 hashes (160 bytes) regardless of the number of transactions in the network.

Address Derivation Function: BLAKE3(Ed25519 public key) → 256 bits → first 23 bits as slot index. Collision probability with 1 million users is less than $10^{-3}$ (birthday paradox over 8M slots).Slot Collision Resolution: If two users obtain the same index, linear probing is used — the next free slot in memory. The wallet stores the final offset in a local keychain file.

## Actum Format (Transaction)

An Actum is a binary structure with a fixed length of 128 bytes:

```
[ 0..31]  Sender_PubKey     Ed25519 public key        (32 B)
[32..54]  Receiver_Slot     uint32 + 3B padding       (4+3 B)
[55..62]  Amount_TOL        uint64                    (8 B)
[63..70]  Fee_TOL           uint64                    (8 B)
[71..78]  Nonce             uint64                    (8 B)
[79..82]  Chronos_Ticket    uint32 (block number)     (4 B)
[83..83]  Expiry_Blocks     uint8 (max 255)           (1 B)
[84..147] Signature         Ed25519                   (64 B)
```

The signature covers bytes [0..83] — all fields except the signature itself. The fixed length of 128 B allows O(1) access to any transaction in the Manifest without parsing.

## 64 MB Header Structure

The Header has three sections with fixed offsets to allow direct access without parsing:

**Section A** [0 MB .. 16 MB]: Current Epoch — 2,100,000 blocks × 8 B (BLAKE3 hash truncated to 8 B) = 16.8 MB (offset of Section A is 0, Section B begins at offset 16,777,216 B)

**Section B** [16 MB .. 48 MB]: Master Hash Registry — Max. 1,000,000 epochs × 32 B (full BLAKE3) = 32 MB (covers 4,000,000 years of operation — practically infinite)

**Section C** [48 MB .. 64 MB]: System Data (16 MB) — Network rules, difficulty setting R, TPM registry, Bitrate calibration

**Section D** (reserved for future network expansion, see document Expansion of Bohemiancoin) will be defined separately upon activation of the first expansion phase.

# Security Analysis and Potential Threats

### Fork — Two Competing Versions of the Codex

Threat: Two Hunters seal closely timed blocks that spread through the network simultaneously. Part of the network accepts block A, the rest block B. In the next round, nodes work on different Codexi.

Solution: The Precision Rule (smaller distance d wins) is a well-chosen deterministic tiebreaker. However, it must be supplemented with an explicit 'wheat path' rule (chain selection): nodes always switch to the branch with the lowest cumulative distance (sum d of all blocks in the branch), not the longest chain as in Bitcoin. This is appropriate because Bohemiancoin does not have a classic chain — it has a sequence of Codexi. Cumulative precision serves an equivalent role. Transition window: nodes accept an alternative block for a maximum of 5 blocks. After this window, the branch with greater cumulative precision is canonical.

### Codex Transition During Fork (Pendens Safety Net)

Threat: If a node switches from branch A to branch B, its Pendens records may not match the canonical Codex of branch B.

Solution: When switching branches, nodes perform a Pendens Rewind — sequential cancellation of all Pendens states recorded in the detached blocks. Algorithm: (1) Identify the common previous Codex (checkpoint). (2) Traverse the delta-logs of blocks A1..An in reverse order and restore original slot values. (3) Apply deltas B1..Bm. Maximum rewind depth is 5 blocks (see transition window), meaning a maximum of 30 × 5 = 150 affected slots.

### Double Spend After Codex Closing

Threat: An attacker prepares a transaction with a low Chronos Ticket (old block number) that technically appears valid, but targets already spent funds.
Solution: The Virbitirium rejects any Actum where Chronos_Ticket < (current_block - MAX_TICKET_AGE). The recommended value of MAX_TICKET_AGE = 3 blocks. This

guarantees a minimum buffer for users with a weak connection, while eliminating attacks with old tickets.

## Sybil Attack on the Virbitirium (Transaction Spam)

Threat: An attacker generates a large number of valid addresses and floods the Virbitirium with micro-transactions with low fees, thereby blocking patient users.

Solution: Three-layer protection. First layer: The Chronos Ticket requires a 3–5 second computation per transaction, reducing throughput to a maximum of approximately 15 transactions/s per computer. Second layer: The Virbitirium has a global size limit (recommended 50,000 pending transactions); when exceeded, new transactions are accepted only if their fee exceeds the median of the current queue. Third layer: the Active status bit in the slot — an address with no balance cannot send a valid Actum, as validation in the Codex will prove a zero balance.

## Attack on the Master Hash Epoch (51% in Grand Strike)

Threat: The Grand Strike occurs once every 4 years and rewards 140,000 GRS to one Hunter. This creates an extreme incentive for concentrating power just before the epoch closes.

Solution: The proposal is to extend the reward with a temporal distribution (Epoch Drip): instead of one large payment of 140,000 GRS, the winner receives only 14,000 GRS immediately, with the remainder (126,000 GRS) paid out linearly over the following 10,000 blocks, approximately 7 days. If the Hunter stops validating during this time, the unpaid portion is redistributed into the Drip fund of the next epoch. This discourages short 'pump and dump' power attacks and encourages active participation across multiple block cycles.

## Loss of CBOR Proof and Auditability

Threat: If a user loses their .cbor file (fire, disk failure), they have no way to prove an earlier transaction. The network itself does not hold history.

Solution: This is an intentional design — personal responsibility is the philosophy of Bohemiancoin. However, we propose formalising three layers of custody: (1) Self-custody — local storage, full control. (2) Archive nodes — voluntary nodes storing historical CBOR packages for a fee (does not change consensus, purely an archival service). (3) Social recovery — upon setting up a wallet, the user can designate 3 of 5 trusted contacts to whom they encrypt a key backup (Shamir's Secret Sharing). None of them has the entire key, but any 3 together can reconstruct it.

## Security Matrix Summary

The following table summarises threats, their assessment, and proposed measures:

| Threat | Impact | Solution |
|---|---|---|
| Fork / race condition | HIGH | Cumulative Precision Rule + Pendens Rewind (5 blocks) |
| Software latency emulation | HIGH | Sequential Dependency Chain + PoPR breadcrumbs |
| Double spend (old ticket) | HIGH | MAX_TICKET_AGE = 3 blocks |
| Virbitirium spam | MEDIUM | Three-layer protection (Chronos + limit + zero balance) |
| Grand Strike concentration | MEDIUM | Epoch Drip (linear payout over 7 days) |
| RAM disk farm | MEDIUM | Codex Salt per-block + Sequential Dep. Chain |
| Loss of CBOR proof | PERSONAL | Shamir's Secret Sharing + Archive nodes |

**Network Expansion: Bohemiancoin is designed so that the capacity limit of a single Codex is not a ceiling for community growth. The Sharded Universe protocol, described in a separate document Expansion of Bohemiancoin, defines a method of scaling without increasing hardware requirements for Hunters.**